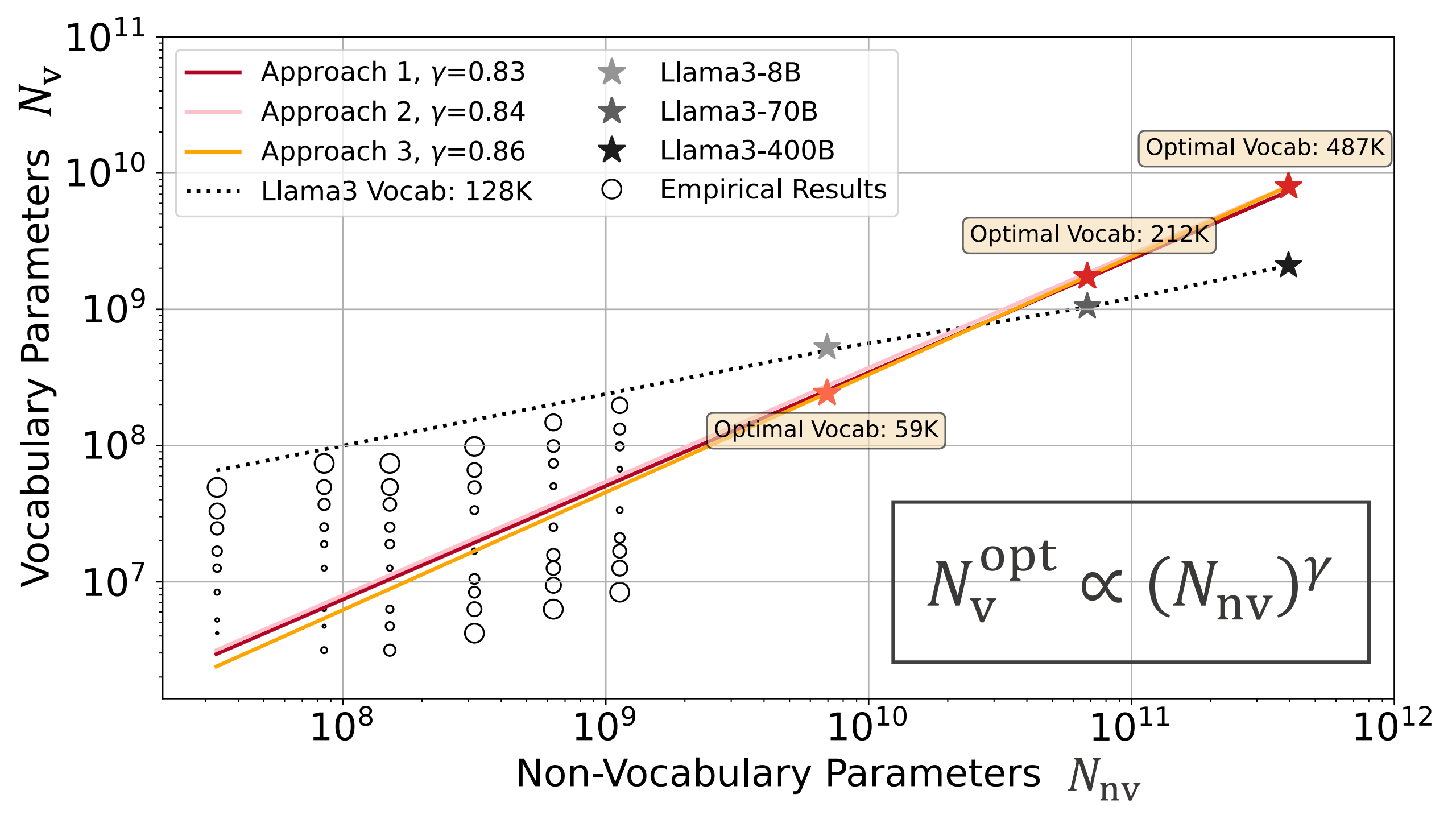
Approach 1, γ=0.83
Approach 2, γ=0.84
Approach 3, γ=0.86
Llama3 Vocab: 128K
Llama3-8B
Llama3-70B
Llama3-400B
Empirical Results
Optimal Vocab: 487K
Optimal Vocab: 212K
Optimal Vocab: 59K
$N_v^{opt} \propto (N_{nv})^{\gamma}$
Vocabulary Parameters N_v
Non-Vocabulary Parameters N_{nv}
10^{7}
10^{8}
10^{9}
10^{10}
10^{11}
10^{12}